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Entangling atomic spins with a Rydberg-dressed spin-flip blockade

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Controlling the quantum entanglement between parts of a many-body system is key to unlocking the power of quantum technologies such as quantum computation, high-precision sensing, and the simulation of many-body physics. The spin degrees of freedom of ultracold neutral atoms in their ground electronic state provide a natural platform for such applications thanks to their long coherence times and the ability to control them with magneto-optical fields. However, the creation of strong coherent coupling between spins has been challenging. Here we demonstrate a strong and tunable Rydberg-dressed interaction between spins of individually trapped caesium atoms with energy shifts of order 1 MHz in units of Planck's constant. This interaction leads to a ground-state spin-flip blockade, whereby simultaneous hyperfine spin flips of two atoms are inhibited owing to their mutual interaction. We employ this spin-flip blockade to rapidly produce single-step Bell-state entanglement between two atoms with a fidelity ≥81(2)%.

Pristine quantum control of many-body correlations is fundamental to realizing the power of quantum information processors. Steady progress has continued in various platforms ranging from solid-state spintronics[1] and superconductors[2,3] to nanophotonics[4] and ultracold trapped atoms, both ionic[5–7] and neutral[8–10]. Cold neutral atoms are particularly attractive as the ability to create entanglement between atoms would allow for greatly increased precision of interferometers for applications in clocks[11–13], and force sensors[14–16]. In addition, cold atoms provide a natural platform for quantum simulation of condensed-matter physics[17,18] and scalable digital quantum computers[19–21]. Controlled entanglement of neutral atoms, however, has been challenging, particularly if one seeks tunable interactions that are strong, coherent and long-range (∼µm).

One mechanism to achieve strong, long-range coupling is the Rydberg blockade[22]. This has been successfully employed for implementing controlled entangling interactions between atoms[9,10,23] and quantum logic gates[24]. In the standard protocol, short pulses excite the population of one atom to the Rydberg state and optical excitation of a second atom is blockaded because of the electric dipole–dipole interaction[21] (EDDI). An alternative protocol is to adiabatically dress the ground state with the excited Rydberg state[25–27]. This Rydberg-dressed interaction enables tunable, anisotropic interactions that open the door to quantum simulations of a variety of exotic quantum phases[26,28,29]. In addition, it allows for quantum control of interacting atoms based solely on microwave/radiofrequency fields whose phase coherence is easily maintained. Applications include spin-squeezing for metrology[13,25], and quantum computing[30,31]. Although the promise of Rydberg-dressed interactions is great, experimental demonstration has been elusive. We present here a clear measurement of this interaction between two Rydberg-dressed atoms and employ coherent control in the ground-state manifold to create entangled Bell states.

Rydberg dressing arises from the modification of the light shift (that is, the optical a.c. Stark shift) of two atoms due to the EDDI. We characterize this by the interaction strength $J \equiv \Delta E_{\rm LS}^{(2)} - 2\Delta E_{\rm LS}^{(1)}$, where $\Delta E_{\rm LS}^{(2)}$ is the light shift for two atoms in the presence of EDDI and $\Delta E_{\rm LS}^{(1)}$ is the light shift for a lone atom optically excited near a Rydberg level. By tuning the Rydberg excitation laser near the Rydberg resonance, double excitation of two Rydberg atoms is blockaded when $U_{\rm dd} \gg \hbar\Omega_{\rm L}$, where $\Omega_{\rm L}$ is the laser Rabi frequency and $U_{\rm dd}$ is the EDDI energy, which scales from $1/R^6$ to $1/R^3$ depending on the interatomic distance R. The resulting difference in the two-atom spectrum implies that $\Delta E_{\rm LS}^{(2)} \neq 2\Delta E_{\rm LS}^{(1)}$, which plateaus at $J = \hbar/2[\Delta_{\rm L} + {\rm sign}(\Delta_{\rm L})(\sqrt{\Delta_{\rm L}^2 + 2\Omega_{\rm L}^2} - 2\sqrt{\Delta_{\rm L}^2 + \Omega_{\rm L}^2})]$, where $\Delta_{\rm L}$ is the laser detuning, when the blockade is perfect[26]. In the strong dressing regime, this can lead to a very strong interaction, $J \sim \hbar\Omega_{\rm L}$, when $\Omega_{\rm L} \sim \Delta_{\rm L}$.

We employ ^{133}Cs atoms, and encode qubits in the hyperfine 'clock states' $|0\rangle \equiv |6S_{1/2}, F=4, m_{\rm F}=0\rangle$, $|1\rangle \equiv |6S_{1/2}, F=3, m_{\rm F}=0\rangle$, of two atoms individually trapped in optical tweezers at 938 nm and separated by a few micrometres[23]. By choosing $|\Delta_{\rm L}|$ small compared with the 9.2 GHz hyperfine splitting, in the presence of EDDI, only the two-qubit state, $|0,0\rangle$, receives a non-negligible, two-atom dressing energy J. As this energy is a shift of the Rydberg-dressed ground state, it can be used to control two-body interactions solely with microwave-frequency fields. In particular, by applying 9.2 GHz radiation on the clock-state resonance (for example, using two-photon Raman lasers) with Raman–Rabi frequency $\Omega_{\rm mw}$, two atoms initially in the state $|1\rangle$ are blockaded from undergoing double spin-flips, $|1,1\rangle \rightarrow |0,0\rangle$, when $J/\hbar \gg \Omega_{\rm mw}$. This is a spin-flip blockade (Fig. 2a) in the dressed ground state, analogous to the familiar excited-state Rydberg blockade.

Note, unlike the single-atom light shift, the ratio of dressing energy J to the photon scattering rate (the fundamental source of decoherence) improves closer to resonance and decreases monotonically with increasing $\Delta_{\rm L}$ at very small detuning. In particular, far from resonance $J \sim \Delta_{\rm L}^{-3}$, whereas absorption followed by spontaneous decay scales as $\gamma \sim \Delta_{\rm L}^{-2}$. For this reason, we operate in a regime of strong Rydberg dressing with a detuning, $\Delta_{\rm L} \leq \Omega_{\rm L}$. For example, one of the experimental conditions used in Fig. 2 is $\Omega_{\rm L}/2\pi = 4.3$ MHz and $\Delta_{\rm L}/2\pi = 1.3$ MHz, where the dressed state has a strong admixture of Rydberg character, $0.6|r\rangle + 0.8|0\rangle$, or 64% probability in $|0\rangle$. Although there is about 36% probability in $|r\rangle$

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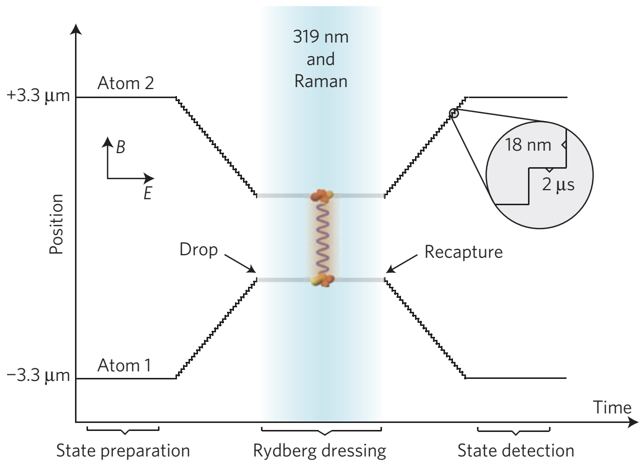

Figure 1 | Experiment sequence. To achieve both a strong ground-state atom–atom interaction and high-fidelity signal detection we perform the following steps at different interatomic spacings. In the experimental procedure, two Cs atoms are initially 6.6 μm apart, and held by optical tweezers. After qubit-state preparation, the two trapped atoms translate towards each other with an average speed of 9 mm s^{-1} (18 nm step every 2 μs) by ramping the modulation frequencies of the AOM. At the target distance, the Rydberg-dressing laser at 319 nm turns on to illuminate the two atoms simultaneously with a Raman laser. The tweezers are extinguished during this step to eliminate optical perturbation. The two atoms then translate back to the original positions for state detection.

of the dressed state, the light shift on the Rydberg-dressed state is insensitive to thermal motion that gives rises to a fluctuation in the difference in the optical phase seen at the relative positions between the two atoms. Such thermal noise was a limiting factor in the generation of spin entanglement in previous work[10]. The thermal atomic motion in the Rydberg-dressed interactions, however, does lead to a Doppler shift, and thus noise in the optical detuning Δ_L.

The key mechanism that determines the interaction strength, J, depends on the EDDI and the optical Rabi frequency Ω_L. Thus, the distance between the two Rydberg-dressed atoms and the choice of the principal quantum number are crucial experimental parameters. Ideally, we would like the atoms to be located far apart for individual addressing, and conversely, in close proximity to maximize J. Our particular implementation of an acousto-optic modulator (AOM) allows us to create two optical tweezers that trap each atom from the same laser by simultaneously driving the AOM at two frequencies (see Supplementary Methods). We achieve this goal by independently sweeping the values of these frequencies and dynamically translating the traps. The capability of producing sufficiently strong J at shorter interatomic distance allows us to reduce the principal quantum number of the Rydberg level. Thus, the sensitivity to external fields, which rapidly increases for high-lying Rydberg levels and is a common challenge in these experiments, is reduced.

Results

We directly measure J as a function of the interatomic distance. Our experiment is illustrated in Fig. 1. The two trapped ^{133}Cs atoms are initially prepared in state $|1,1\rangle$ and we dynamically translate them to be in close proximity at a targeted distance R. We then extinguish the tweezers for a short time to eliminate light shifts from the dipole-trap laser, and immediately apply a short pulse of the Raman and Rydberg-dressing lasers concurrently. Afterwards, the optical tweezers are restored to recapture the falling atoms and translate them back to the original positions for independent state detection, which is accomplished by using the $|6S_{1/2}, F=4\rangle \rightarrow |6P_{3/2}, F'=5\rangle$ D2 cycling transition to determine whether each atom is in state $|0\rangle$ (bright to this excitation) or $|1\rangle$ (dark to this excitation). We use a 319-nm laser for dressing the ^{133}Cs atom, which couples atoms directly from the ground state to the Rydberg level, $6S_{1/2}, F=4 \rightarrow 64P_{3/2}$, in a single-photon transition[23]. This avoids unwanted population in an intermediate, short-lived excited state that arises in the typical two-photon Rydberg excitation method, which causes additional ground-state decoherence[30]. We choose a detuning that is small compared with the ground-state hyperfine splitting so that the dressing of $|1,1\rangle$ in the $F=3$ manifold is negligible, but all other ground states in the logical basis, $\{|0,0\rangle, |0,1\rangle, |1,0\rangle\}$, are now well described in the dressed basis. To drive spin flips, we apply the Raman laser fields to the two Rydberg-dressed atoms when they are at a desired separation. By sweeping the Raman (microwave) frequency and measuring the

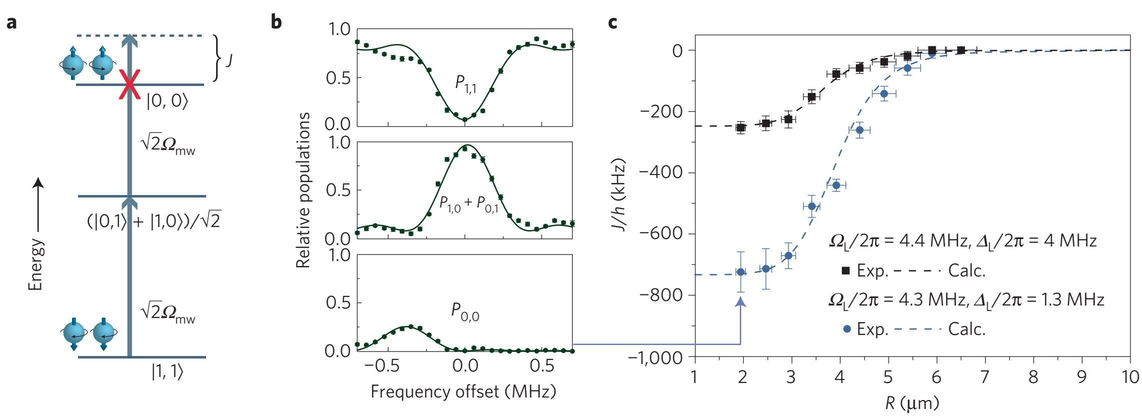

Figure 2 | Rydberg-dressed ground-state interaction J and the spin-flip blockade. a, Energy-level diagram of the spin-flip blockade on the Rydberg-dressed two-qubit sublevels. For a sufficiently large J, only the transition from $|1,1\rangle \rightarrow (|1,0\rangle$ or $|0,1\rangle)$ is allowed and the double spin-flip transition from $(|1,0\rangle$ or $|0,1\rangle) \rightarrow |0,0\rangle$ is blockaded when microwave radiation (stimulated Raman transition in our case) is applied at the non-interacting, single-atom qubit resonance frequency. **b**, Scanning the microwave frequency of the stimulated Raman pulse applied to the two trapped Rydberg-dressed ^{133}Cs atoms reveals the ground-state spin-flip blockade. The excitation from $|1,1\rangle \rightarrow |0,0\rangle$ occurs through an anti-blockade two-photon transition. J/h is simply twice the shift of the resonance frequency for excitation to the state $|0,0\rangle$. **c**, Experimental data of J versus R with two sets of parameters. The dashed curves are the calculated values based on a detailed model with no free parameters (see Supplementary Methods). The error bars shown in **b** and **c** correspond to one standard deviation.

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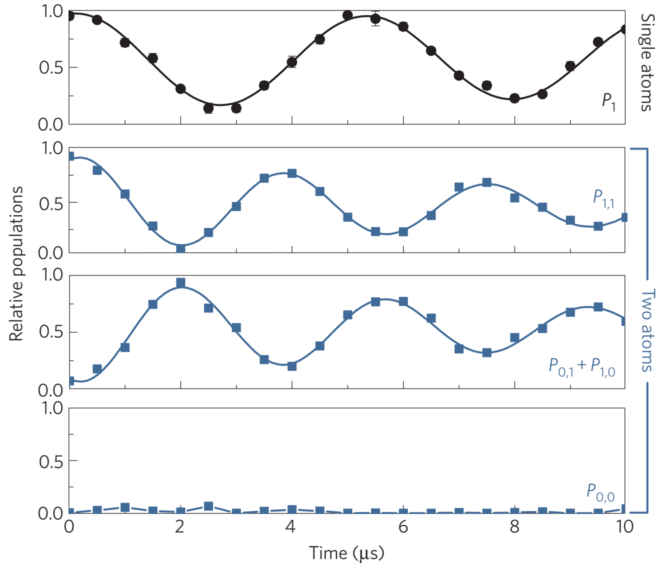

Figure 3 | Generating entanglement directly. Top panel: Rabi oscillations of a single Rydberg-dressed Cs qubit. Lower three panels: two-atom data with Rydberg-dressed spin-flip blockade ($J/h\approx750$ kHz with $\Omega_L/2\pi=4.3$ MHz, $\Delta_L/2\pi=1.1$ MHz, $R=2.9$ µm). The data points are fitted with curves of damped oscillation and exponentially varied offset. Rabi oscillation occurs between two spin-down atoms and a two-qubit entangled state. There is a $\sqrt{2}$ enhancement of the microwave Rabi rate, Ω_{mw}, arising from the blockade, and excitation to state $|0,0\rangle$ is strongly suppressed owing to the transition blockade as shown in Fig. 2a. The maximum Bell state $|\Psi_+\rangle$ entanglement is thus generated at around 2 µs. The error bars for all data points correspond to one standard deviation.

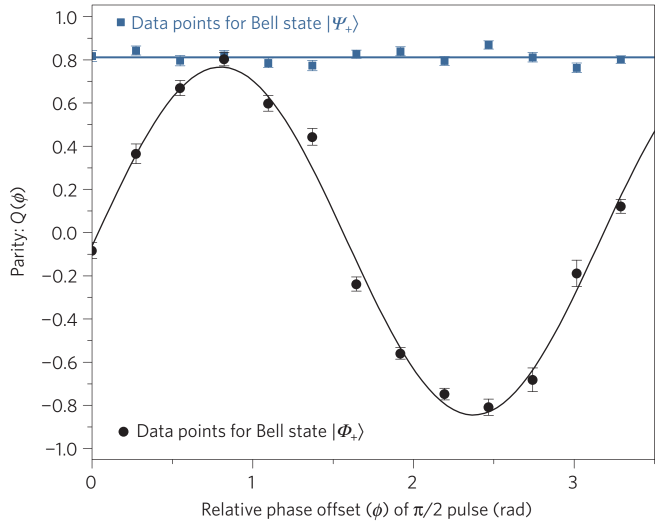

Figure 4 | Entanglement verification. A global $\pi/2$ pulse is applied to the undressed system after the entangled state is prepared. The data are obtained with the same experimental parameters used for data in Fig. 3. $|\Psi_+\rangle$ data are fitted with a straight line, and $|\Phi_+\rangle$ data are fitted with a sinusoidal function. It shows that both Bell states generated from our experiment have a fidelity $\geq$81(2)%. Here, ρ represents the two-qubit density matrix. The parity measurement, $P_{1,1}+P_{0,0}-(P_{0,1}+P_{1,0})$, allows direct determination of the amplitudes of the off-diagonal elements for both entangled states. The error bars for all data points correspond to one standard deviation.

resulting spin flips, we obtain a two-qubit energy spectrum of the form shown in Fig. 2b. To determine J, we measure the microwave resonance frequency for the transition $|1,1\rangle\rightarrow(|1,0\rangle+|0,1\rangle)/\sqrt{2}$ and for the two-microwave-photon transition $|1,1\rangle\rightarrow|0,0\rangle$. $J/\hbar$ is given by twice the frequency difference of these two resonances. The microwave Rabi rate Ω_{mw} and the pulse time of the stimulated Raman laser are properly chosen to ensure the observation of both the two-photon and the single-photon microwave resonance. The relative populations $P_{1,1}$, $P_{1,0}$, $P_{0,1}$ and $P_{0,0}$ of the four two-qubit computational basis states can be directly measured with the coincident bright and dark signals determined by the photon counting of the two avalanche photodiodes. Figure 2c plots the measured J as a function of interatomic distance for two different combinations of the 319-nm laser detuning Δ_L and the optical Rabi rate Ω_L.

Although the simple two-level atom model gives a clean theoretical prediction for J as described above, the true atomic physics is more complex. The EDDI shifts $64P_{3/2}$ negatively (red); thus, for Rydberg dressing we tune the 319-nm laser to the blue. At short interatomic spacings, however, the two-atom Rydberg energy levels strongly mix to yield a spectrum with molecular quality[32]. The resulting 'spaghetti' of molecular levels could potentially lead to additional unwanted resonances that would ruin the Rydberg blockade. This may also affect the lifetime of the two-atom dressed state. The existence of such resonances, however, depends on the oscillator strengths. With our experimental resolution, we are unable to detect such resonances in our experiment. We use a best-fit of the blockade shift curve of $|r,r\rangle$ from our detailed model[30] to calculate J as shown in Fig. 2c (see Supplementary Methods and Supplementary Fig. 3). With an ideal Rydberg blockade, $U_{dd}\rightarrow\infty$, we find the expected plateau in J at short interatomic distances predicted from the two-level model. The plateau at small R is characteristic of a perfect Rydberg blockade, and agrees with simple theoretical predictions.

With large values of J, we can employ the spin-flip blockade to create entanglement between atomic spin qubits. By driving a resonant Raman pulse $|1\rangle\rightarrow|0\rangle$ simultaneously on the two atoms, we cause Rabi oscillations between $|1,1\rangle$ and the entangled Bell state, $|\Psi_+\rangle=(|0,1\rangle+|1,0\rangle)/\sqrt{2}$, as the experimental data show in Fig. 3. A signature of the spin-flip blockade is the characteristic increase in the Raman–Rabi frequency to $\sqrt{2}\Omega_{mw}$ (ref. 10). The Bell state $|\Phi_+\rangle=(|0,0\rangle+|1,1\rangle)/\sqrt{2}$, the two-atom cat state, can be generated from $|\Psi_+\rangle$ by subsequently applying a global $\pi/2$ rotation on the qubits. Alternatively, while the Rydberg-dressing laser is still on, a two-microwave-photon $\pi/2$-pulse at a shifted frequency, resonant with the transition from $|1,1\rangle\rightarrow|0,0\rangle$, also generates $|\Phi_+\rangle$. This method has a lower fidelity because the two-photon microwave Rabi rate must be small to avoid excitation to the off-resonant $|\Psi_+\rangle$ state, and decoherence is more likely on this long timescale.

We measure the fidelity of Bell-state preparation as follows. For a given J, the Raman pulse duration T is chosen so that $\sqrt{2}\Omega_{mw}T=\pi$ with $\Omega_{mw}\ll J/\hbar$. Following this procedure, the automated experimental control system checks whether both atoms are still present in the traps; if so, it counts as a 'valid' operation. We determine the lower bound of the entanglement fidelity by measuring the off-diagonal coherence between the two-qubit logical basis states, $\langle x',y'|\rho|x,y\rangle$, where ρ is the two-qubit density matrix. For this, we apply a global $\pi/2$ pulse with phase ϕ to the entangled state. As a function of ϕ, we measure the expected value of the parity $Q(\phi)=\langle\sigma_z\otimes\sigma_z\rangle_\phi=[P_{1,1}+P_{0,0}-(P_{0,1}+P_{1,0})](\phi)$, where $P_{x,y}(\phi)$ is the population in the logical state $|x,y\rangle$ after application of the $\pi/2$ pulse[33,34]. For qubits prepared in the $|\Psi_+\rangle$ state, Q is independent of ϕ and always a positive number; $Q=1$ when the entanglement is perfect. For qubits prepared in the $|\Phi_+\rangle$ state, $Q(\phi)$ is an oscillating function of ϕ. In this case, perfect entanglement corresponds to perfect oscillation visibility. The entanglement fidelity (fidelity between the prepared state and the target Bell state) is $\geq2|\langle x',y'|\rho|x,y\rangle|$, as measured from $Q(\phi)$ (ref. 34). When this fidelity is greater than 50%, the state is necessarily entangled. The measurement of $Q(\phi)$ in Fig. 4 shows that with the same

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experimental parameters as used in Fig. 3, we have achieved entanglement fidelity $\geq 81 \pm 2\%$ for generating both $|\Psi_+\rangle$ and $|\Phi_+\rangle$ when both atoms are recaptured in the trap.

Discussion

Given that we measure the two-atom survival probability after the procedure to be $74 \pm 2\%$, the success rate of deterministically generating an entangled qubit pair is therefore $\geq 60 \pm 3\%$. With our current experimental data rate ($\approx 10\,\mathrm{s}^{-1}$), on average we generate 6 pairs of entangled qubits per second. The current factors limiting the entanglement fidelity given a valid procedure are: the optical pumping efficiency, decay of the Raman Rabi oscillation of the Rydberg-dressed states (Fig. 3), and the strength of J. Optical pumping efficiency determines how well we can prepare the atoms in the qubit subspace (computational space), which can be improved with a more careful pumping scheme. Decay of the Rabi oscillation impacts the fidelity of the π pulse; we can improve this fidelity by increasing Ω_{mw} but only for sufficiently large J. One identified contribution to the decay is the shorter than expected single-atom Rydberg-state lifetime (see Supplementary Methods). Overall, a comprehensive explanation of the two-atom Rabi oscillation decay is the subject of future work.

The protocol presented here is quite robust despite some technical imperfections, and we expect improvement in future experiments. For example, at present the Bell-state entanglement is generated in the dressed basis, resulting in atom loss due to the admixture of the Rydberg state. In principle dressed states can be returned to the bare ground states by adiabatically ramping the dressing laser's intensity and detuning in an optimal way[31], something we will implement in a future generation of the experiment. Finally, minimizing other experimental imperfections will most likely improve the Rydberg-state lifetime, improve the deterministic entanglement fidelity, and open the door to more full control of complex quantum systems.

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Author contributions

Experimental work by Y.-Y.J., A.M.H. and G.W.B. Numerical modelling by Y.-Y.J. Theoretical work by I.H.D. and T.K.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to G.W.B.

Competing financial interests

The authors declare no competing financial interests.

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